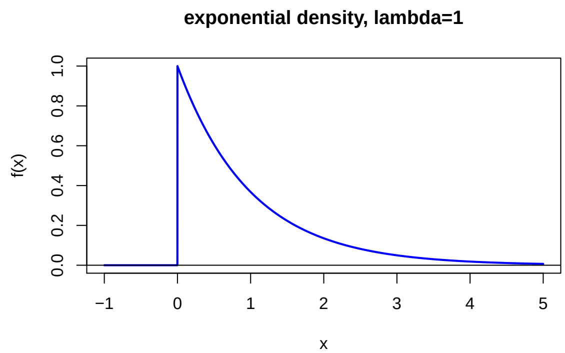
Solution exercise set 2

Problem 1:

a)



b)

$$
\begin{aligned}
\mathrm{E}(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x\lambda e^{-\lambda x} dx = [-xe^{-\lambda x}]_0^{\infty} - \int_0^{\infty} -e^{-\lambda x} dx \\
&= 0 - [(1/\lambda)e^{-\lambda y}]_0^{\infty} = \underline{\underline{1/\lambda}} \\
\mathrm{E}(X^2) &= \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx = [-x^2 e^{-\lambda x}]_0^{\infty} - \int_0^{\infty} 2x(-e^{-\lambda x}) dx \\
&= 0 + 2(1/\lambda) \int_0^{\infty} x\lambda e^{-\lambda x} dx = 2(1/\lambda)^2 \\
\Rightarrow \mathrm{Var}(X) &= \mathrm{E}(X^2) - \mathrm{E}(X)^2 = 2(1/\lambda)^2 - (1/\lambda)^2 = \underline{\underline{1/\lambda^2}}
\end{aligned}
$$

c) First notice that:

$$F(x) = \int_{-\infty}^{x} f(u) du = \int_0^{x} \lambda e^{-\lambda u} du = [-e^{-\lambda u}]_0^x = 1 - e^{-\lambda x}$$

With $\mathrm{E}(X) = 1/\lambda = 10000$ we get $\lambda = 1/10000$ and then:

$$
\begin{aligned}
P(X < 10000) &= F(10000) = 1 - e^{-10000/10000} = \underline{\underline{0.632}} \\
P(X > 5000) &= 1 - PX \leq 50000) = 1 - F(5000) = 1 - (1 - e^{-5000/10000}) = e^{-0.5} = \underline{\underline{0.607}} \\
P(5000 < X < 10000) &= P(X < 10000) - P(X < 5000) = F(10000) - F(5000) \\
&= 1 - e^{-10000/10000} - (1 - e^{-5000/10000}) = e^{-0.5} - e^{-1} = \underline{\underline{0.239}}
\end{aligned}
$$

d) See the accompanying R-code file.

Problem 2:

See the accompanying R-code file for how to find the probabilities in R.

a) With $\mu = \lambda t = 2 \cdot 1$ we get $P(X = x) = \frac{2^x}{x!}e^{-2}$ which gives $P(X = 2) = \frac{2^2}{2!}e^{-2} = \underline{\underline{0.271}}$. Further $P(X = 0) = \frac{2^0}{0!}e^{-2} = 0.135$ and $P(X = 1) = \frac{2^1}{1!}e^{-2} = 0.271$, and thus

$$\begin{aligned} P(X \geq 3) &= 1 - P(X < 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\ &= 1 - 0.135 - 0.271 - 0.271 = \underline{\underline{0.323}}. \end{aligned}$$

b) When we look at a period of 0.5 years we get that the number of accidents is Poisson distributed with $\mu = \lambda t = 2 \cdot 0.5 = 1$, i.e. $P(X = 2) = \frac{1^2}{2!}e^{-1} = \underline{\underline{0.184}}$

c) Over a ten years period the number of accidents is Poisson distributed with $\mu = \lambda t = 2 \cdot 10 = 20$, i.e. $P(X = x) = \frac{20^x}{x!}e^{-20}$. We can use this to calculate $P(X \geq 20) = 1 - P(X \leq 19)$ exactly, but here it is more convenient to use the hint about the approximation to the normal distribution. Since $\mu > 15$ the normal approximation is good, and we get

$$\begin{aligned} P(X \geq 20) &= 1 - P(X \leq 19) = 1 - P(\frac{X - 20}{\sqrt{20}} \leq \frac{19 - 20}{\sqrt{20}}) \\ &= 1 - P(Z \leq -0.22) = 1 - 0.413 = \underline{\underline{0.587}} \end{aligned}$$

In R it is easy to do the exact calculation which gives the result 0.530, see the R-file.

d)

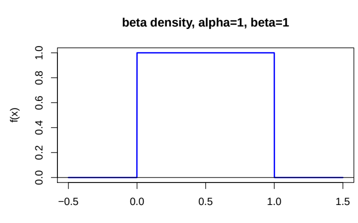
$$P(X > 2 | X \geq 1) = \frac{P(X > 2 \cap X \geq 1)}{P(X \geq 1)} = \frac{P(X > 2)}{1 - P(X = 0)} = \frac{P(X \geq 3)}{1 - 0.135} = \frac{0.323}{0.865} = \underline{\underline{0.373}}.$$

e) See the accompanying R-code file.

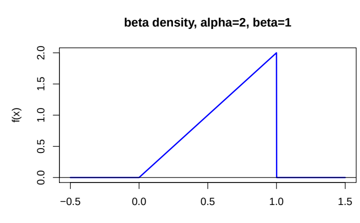
Problem 3:

a)

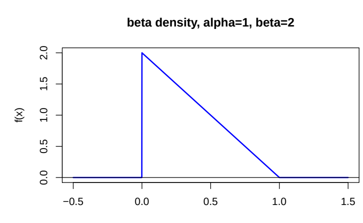
i) $f(x) = 1$



ii) $f(x) = 2x$



iii) $f(x) = 2(1 - x)$



b) See the accompanying R-code solution. When you execute the code many times you should see that the average tends to fall closer to the mean the larger samples you generate.

Problem 4:

a) We have from the lectures that if $U \sim U[a,b]$ then $\mathrm{E}(U) = (a+b)/2$ and $\mathrm{Var}(U) = (b-a)^2/12$. Thus with $a = 0$ and $b = 2$ we get that $\mathrm{E}(U) = (0+2)/2 = \underline{\underline{1}}$ and $\mathrm{Var}(U) = (2-0)^2/12 = \underline{\underline{1/3}}$..

Since $\mathrm{E}(U_1 + \cdots + U_n) = n\mathrm{E}(U_i) = n$ and $\mathrm{Var}(U_1 + \cdots + U_n) = n\mathrm{Var}(U_i)$, the variable:

$$X = \frac{U_1 + \cdots + U_n - n}{\sqrt{n\mathrm{Var}(U_i)}},$$

is standardized (zero mean and unit variance). Further, the central limit theorem says that this variable approaches the (standard) normal distribution when n increases since it is a (standardized) sum of independent, identically distributed (IID) random variables.

b) See the accompanying R-code file.